DHCP Questions

11 questions | Original English questions and source answers

Question 1

When implementing a router as a DHCP server, which two features must be configured? (Choose two)

Options

A. manual bindings

B. database agent

C. smart-relay

D. relay agent information

E. address pool

Answer

A. manual bindings

E. address pool

Source explanation

The following example shows how to configure a DHCP Server on a Cisco router:
Router(config)#ip dhcp pool CLIENTS
Create a DHCP Pool named CLIENTS
Router(dhcp-config)#network 10.1.1.0 /24
Specifies the subnet and mask of the DHCP address pool
Router(dhcp-config)#default-router 10.1.1.1
Set the default gateway of the DHCP Clients
Router(dhcp-config)#dns-server 10.1.1.1
Configure a Domain Name Server (DNS)
Router(dhcp-config)#domain-name 9tut.com
Configure a domain-name
Router(dhcp-config)#lease 0 12
Duration of the lease (the time during which a client computer can use an assigned IP address). The syntax is "lease {days[hours] [minutes] | infinite}". In this case the lease is 12 hours. The default is a one-day lease. Before the lease expires, the client typically needs to renew its address lease assignment with the server.
Router(dhcp-config)#exit
Router(config)# ip dhcp excluded-address 10.1.1.1 10.1.1.10
The IP range that a DHCP Server should not assign to DHCP Clients. Notice this command is configured under global configuration mode.
Manual bindings are IP addresses that have been manually mapped to the MAC addresses of hosts that are found in the DHCP database.
All DHCP clients send a client identifier (DHCP option 61) in the DHCP packet. To configure manual bindings, you must enter the client-identifier DHCP pool configuration command with the appropriate hexadecimal values identifying the DHCP client. For example:
ip dhcp pool SERVER
host 172.16.200.100 255.255.255.0
client-identifier 01aa.bbcc.0003.00
default-router 172.16.200.1
!
Therefore two requirements for DHCP binding is the IP address and the hardware address (MAC address) of the client. Notice that in the above example "aabb.cc00.0300" is the MAC address of the client while prefix "01" represents the Ethernet media type.
Note: In fact, this question should ask "When implementing a router as a DHCP server, which one of these features must be configured?". Because we only need to configure either an "address pool" or "manual binding".

Source: CCNA Training » DHCP Quiz.pdf | Question 1

Question 2

Which command must you enter to configure a DHCP relay?

Options

A. `ip address dhcp`

B. `ip helper-address`

C. `ip dhcp relay`

D. `ip dhcp pool`

Answer

B. ip helper-address

Source explanation

If the DHCP Server is not on the same subnet with the DHCP Client, we need to configure the router on the DHCP client side to act as a DHCP Relay Agent so that it can forward DHCP messages between the DHCP Client & DHCP Server. To make a router a DHCP Relay Agent, simply put the “ip helper-address <IP-address-of-DHCP-Server>” command under the interface that receives the DHCP messages from the DHCP Client.
As we know, router does not forward broadcast packets (it drops them instead) so DHCP messages like DHCPDISCOVER message will be dropped. But with the “ip helper-address ...” command, the router will accept that broadcast message and cover it into a unicast packet and forward it to the DHCP Server. The destination IP address of the unicast packet is taken from the “ip helper-address ...” command.

Source: CCNA Training » DHCP Quiz.pdf | Question 2

Question 3

Refer to the exhibit. If the network environment is operating normally, which type of device must be connected to interface FastEthernet 0/1?

ip arp inspection vlan 2-10
interface fastethernet 0/1
ip arp inspection trust

Options

A. DHCP client

B. PC

C. router

D. access point

Answer

C. router

Source explanation

To configure DHCP snooping feature, at least three steps must be done:
1. Configure global DHCP snooping
Switch(config)# ip dhcp snooping
2. Configure trusted ports (as least on 1 port). By default, all ports are untrusted
Switch(config-if)# ip dhcp snooping trust
3. Configure DHCP snooping for the selected VLANs
Switch(config)# ip dhcp snooping vlan {VLAN-ID | VLAN range}
Note: To configure DHCP snooping with Dynamic ARP Inspection we need to add the command “ip arp inspection vlan vlan-id” in global configuration mode and “ip arp inspection trust” in interface mode.
In a normal network environment, we should trust interfaces that are connected to routers, not end points.

Source: CCNA Training » DHCP Quiz.pdf | Question 3

Question 4

What is a DHCP client?

Options

A. a workstation that requests a domain name associated with its IP address

B. a server that dynamically assigns IP addresses to hosts

C. a router that statically assigns IP addresses to hosts

D. a host that is configured to request an IP address automatically

Answer

D. a host that is configured to request an IP address automatically

Source: CCNA Training » DHCP Quiz.pdf | Question 4

Question 5

Where does a switch maintain DHCP snooping information?

Options

A. in the VLAN database

B. in the CAM table

C. in the DHCP binding database

D. in the MAC address table

Answer

C. in the DHCP binding database

Source: CCNA Training » DHCP Quiz.pdf | Question 5

Question 6

What are two roles of the Dynamic Host Configuration Protocol (DHCP)? (Choose two)

Options

A. The DHCP server leases client IP addresses dynamically

B. The DHCP server offers the ability to exclude specific IP addresses from a pool of IP addresses

C. The DHCP client maintains a pool of IP addresses it can assign

D. The DHCP server assigns IP addresses without requiring the client to renew them

E. The DHCP client can request up to four DNS server addresses

Answer

A. The DHCP server leases client IP addresses dynamically

B. The DHCP server offers the ability to exclude specific IP addresses from a pool of IP addresses

Source: CCNA Training » DHCP Quiz.pdf | Question 6

Question 7

Which type of information resides on a DHCP server?

Options

A. a list of statically assigned MAC addresses

B. a list of the available IP addresses in a pool

C. usernames and passwords for the end users in a domain

D. a list of public IP addresses and their corresponding names

Answer

B. a list of the available IP addresses in a pool

Source: CCNA Training » DHCP Quiz.pdf | Question 7

Question 8

Refer to the exhibit. An engineer deploys a topology in which R1 obtains its IP configuration from DHCP. If the switch and DHCP server configurations are complete and correct. Which two sets of commands must be configured on R1 and R2 to complete the task? (Choose two)



Options

A. R1 (config)# interface fa0/0
R1 (config-if)# ip helper-address 192.0.2.2

B. R1 (config)# interface fa0/0
R1 (config-if)# ip helper-address 198.51.100.100

C. R2(config)# interface gi0/0
R2(config-if)# ip address dhcp

D. R2(config)# interface gi0/0
R2(config-if)# ip helper-address 198.51.100.100

E. R1 (config)# interface fa0/0
R1 (config-if)# ip address dhcp
R1 (config-if)# no shutdown

Answer

D. R2(config)# interface gi0/0
R2(config-if)# ip helper-address 198.51.100.100

E. R1 (config)# interface fa0/0
R1 (config-if)# ip address dhcp
R1 (config-if)# no shutdown

Source: CCNA Training » DHCP Quiz.pdf | Question 8

Question 9

Which Cisco IOS command will indicate that interface GigabitEthernet 0/0 is configured via DHCP?

Options

A. show ip interface GigabitEthernet 0/0

B. show ip interface dhcp

C. show ip interface GigabitEthernet 0/0 dhcp

D. show ip interface GigabitEthernet 0/0 brief

E. show interface GigabitEthernet 0/0

Answer

A. show ip interface GigabitEthernet 0/0

Source: CCNA Training » DHCP Quiz.pdf | Question 9

Question 10

Which command enables a router to become a DHCP client?

Options

A. `ip dhcp pool`

B. `ip helper-address`

C. `ip address dhcp`

D. `ip dhcp client`

Answer

C. ip address dhcp

Source explanation

If we want to get an IP address from the DHCP server on a Cisco device, we can use the command “ip address dhcp”.
Note: The command “ip helper-address” enables a router to become a DHCP Relay Agent.

Source: CCNA Training » DHCP Quiz.pdf | Question 10

Question 11

Where does the configuration reside when a helper address is configured to support DHCP?

Options

A. on every router along the path.

B. on the router closest to the client.

C. on the router closest to the server.

D. on the switch trunk interface.

Answer

B. on the router closest to the client.

Source: CCNA Training » DHCP Quiz.pdf | Question 11